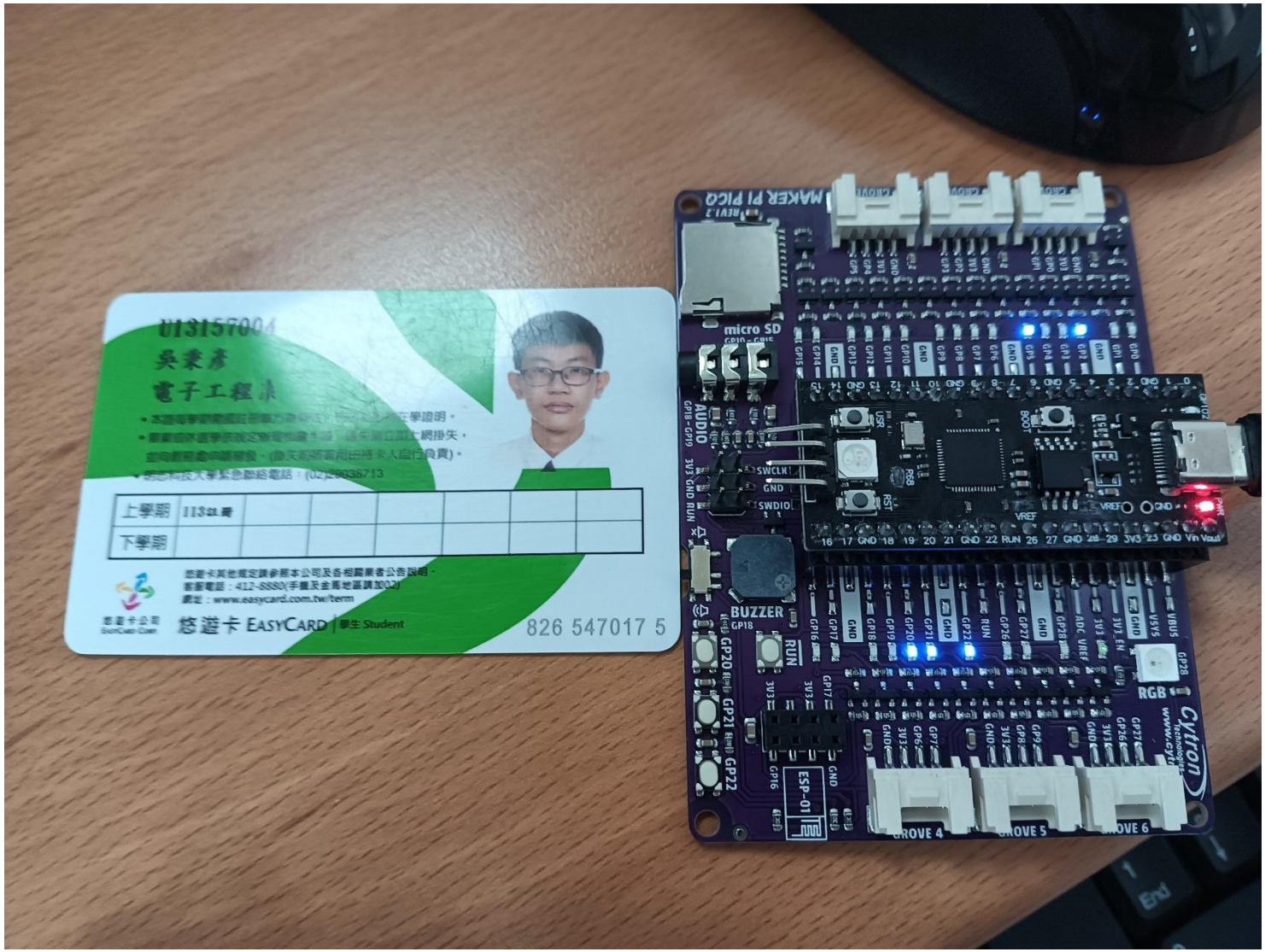
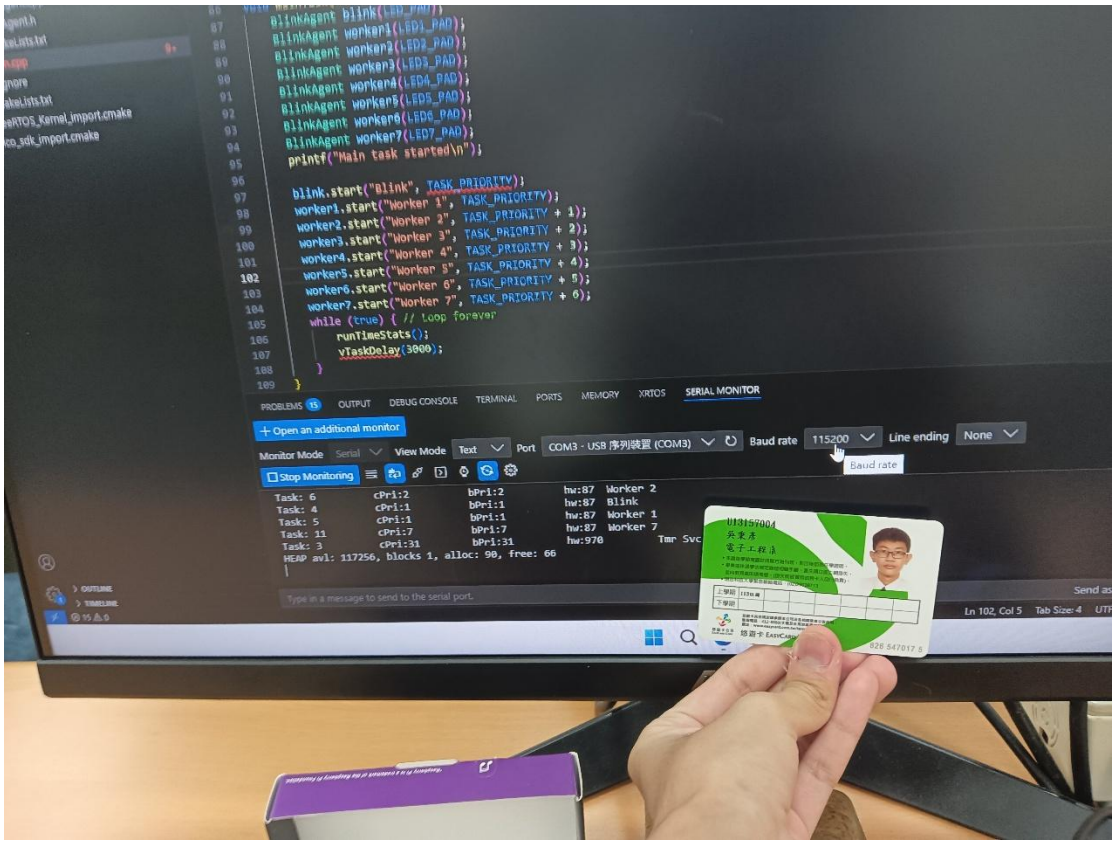
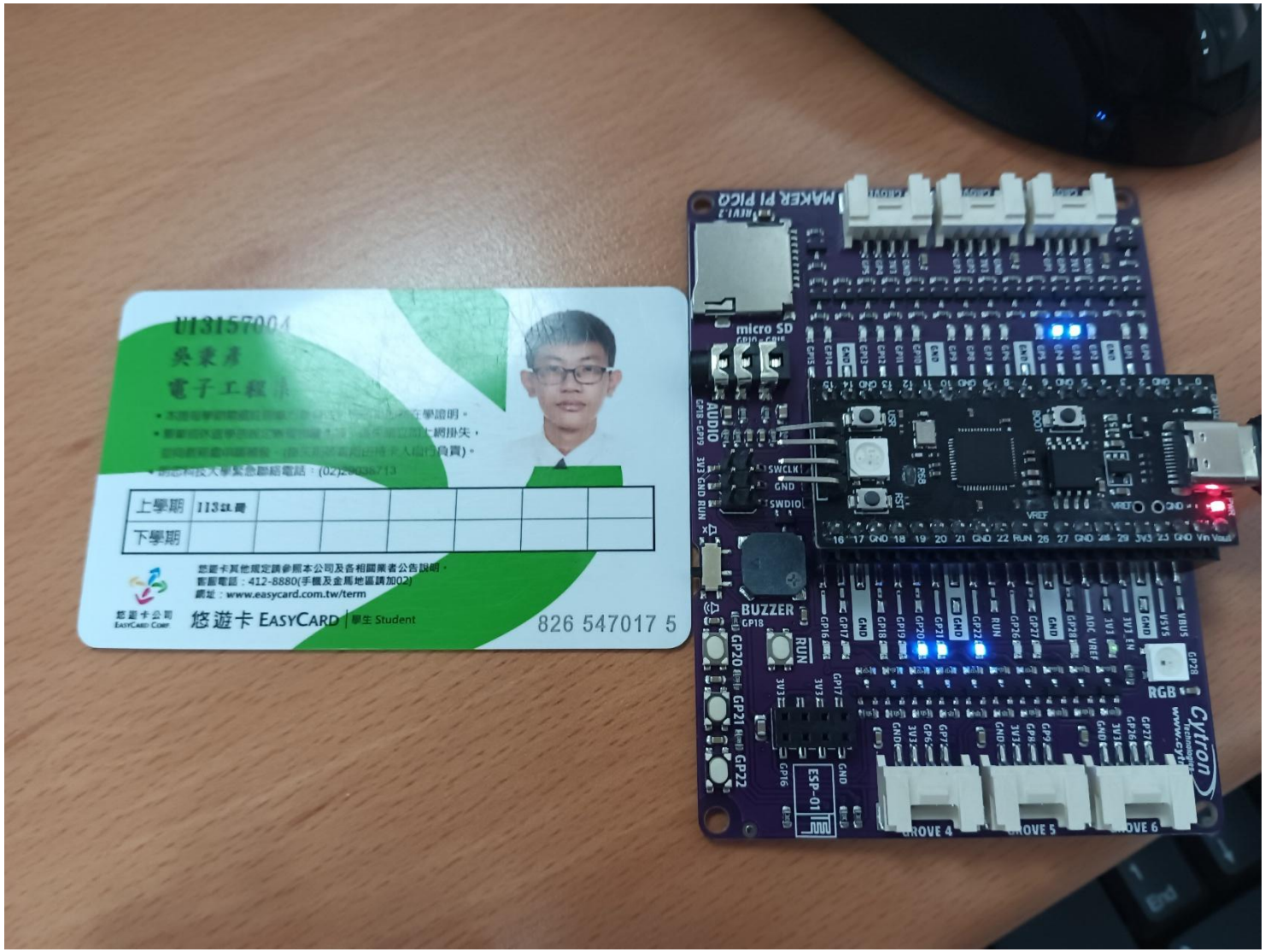
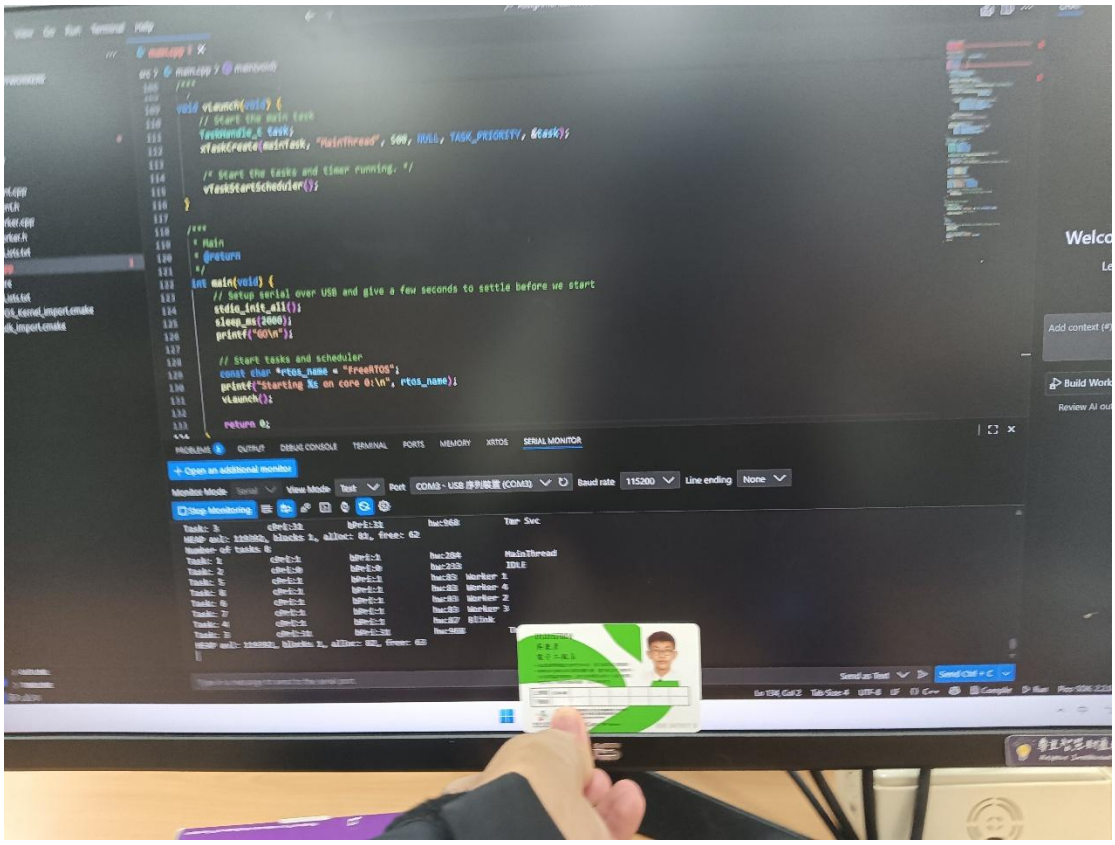
U13157004

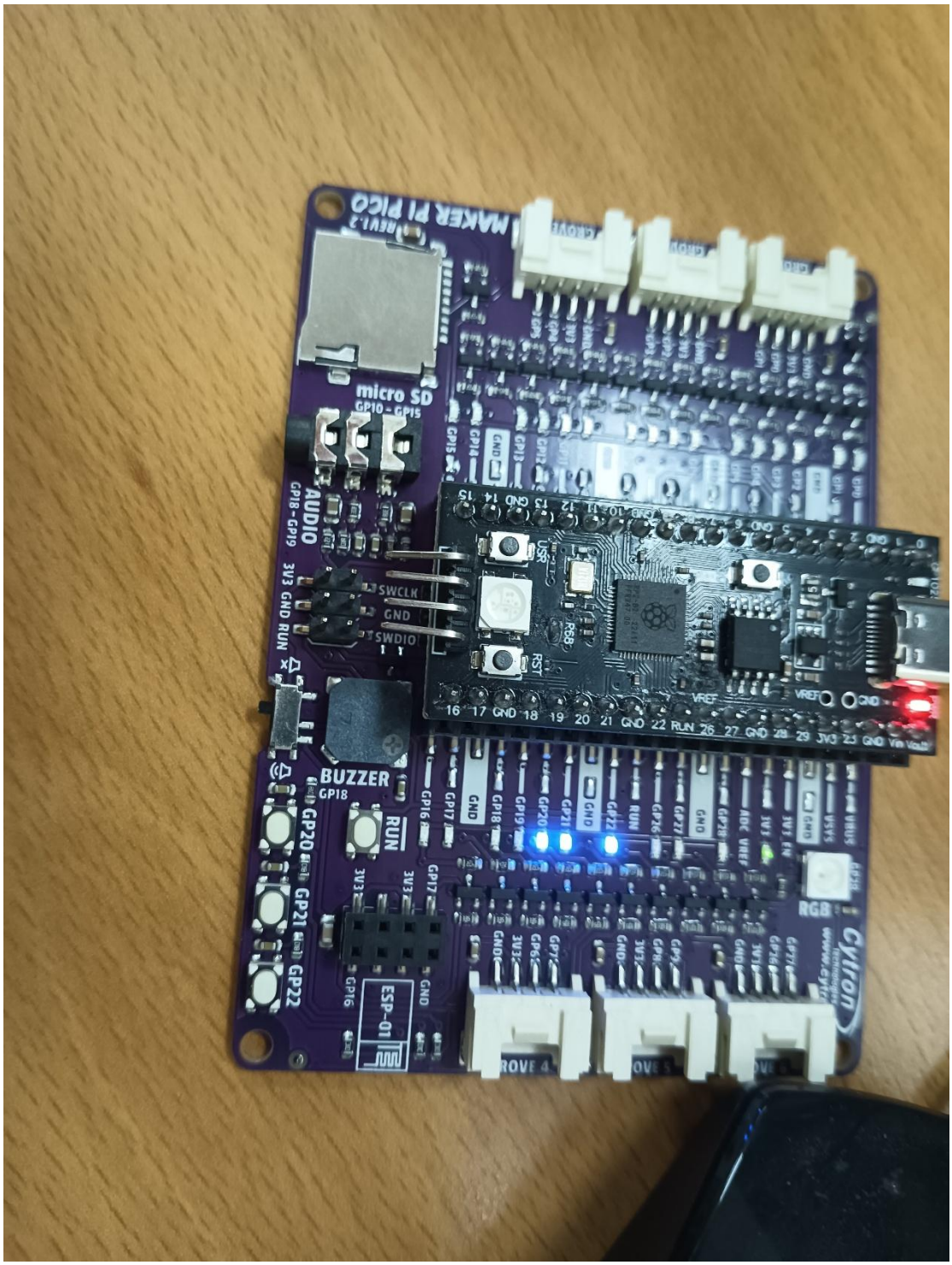
吳秉彥

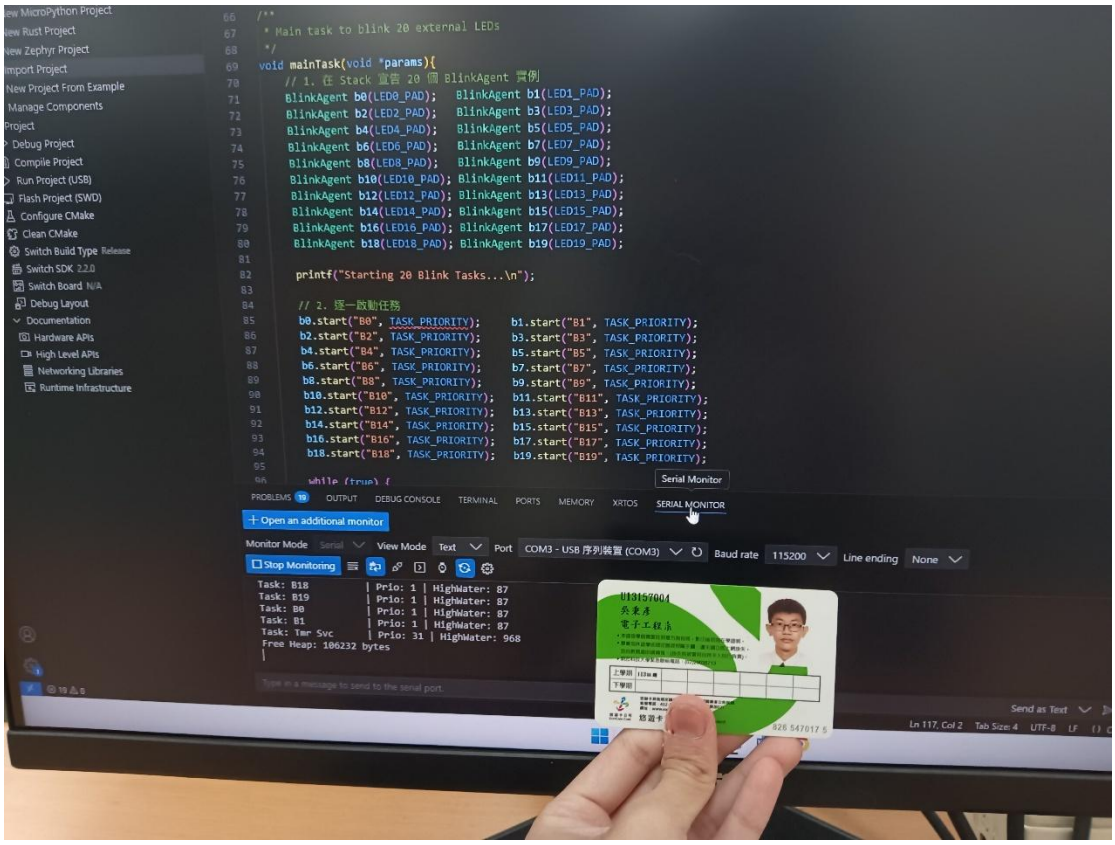

U13157004
吳秉彥
電子工程系
上學期
下學期
826 547017 5
悠遊卡 EASYCARD
MAKER PI PICO
micro SD
AUDIO
BUZZER
RUN
RGB
ESP-01
GROVE 4
GROVE 5
GROVE 6
cytron


BlinkAgent blink(LED_PAD);
BlinkAgent worker1(LED1_PAD);
BlinkAgent worker2(LED2_PAD);
BlinkAgent worker3(LED3_PAD);
BlinkAgent worker4(LED4_PAD);
BlinkAgent worker5(LED5_PAD);
BlinkAgent worker6(LED6_PAD);
BlinkAgent worker7(LED7_PAD);
printf("Main task started\n");
blink.start("Blink", TASK_PRIORITY);
worker1.start("Worker 1", TASK_PRIORITY + 1);
worker2.start("Worker 2", TASK_PRIORITY + 2);
worker3.start("Worker 3", TASK_PRIORITY + 3);
worker4.start("Worker 4", TASK_PRIORITY + 4);
worker5.start("Worker 5", TASK_PRIORITY + 5);
worker6.start("Worker 6", TASK_PRIORITY + 6);
worker7.start("Worker 7", TASK_PRIORITY + 7);
while (true) { // Loop forever
runTimeStats();
vTaskDelay(3000);
}
}
PROBLEMS 15
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
MEMORY
XRTOS
SERIAL MONITOR
Open an additional monitor
Monitor Mode
Serial
View Mode
Text
Port
COM3 - USB 序列裝置 (COM3)
Baud rate
115200
Line ending
None
Baud rate
Stop Monitoring
Task: 6 cPri:2 bPri:2 hw:87 Worker 2
Task: 4 cPri:1 bPri:1 hw:87 Blink
Task: 5 cPri:1 bPri:1 hw:87 Worker 1
Task: 11 cPri:7 bPri:7 hw:87 Worker 7
Task: 3 cPri:31 bPri:31 hw:970 Tmr Svc
HEAP avl: 117256, blocks 1, alloc: 90, free: 66
Type in a message to send to the serial port.
Ln 102, Col 5
Tab Size: 4


U13157004
吳秉彥
電子工程
上學期
113上學
下學期
您遊卡其他規定請參照本公司及各相關業者公告說明。
客服電話：412-8880(手機及金馬地區請加02)
網址：www.easycard.com.tw/term
悠遊卡 EASYCARD | 學生 Student
826 547017 5
MAKER PI PICO
micro SD
AUDIO
BUZZER
RUN
RGB
ESP-01
GROVE 4
GROVE 5
GROVE 6
Cytron


main.cpp
/***
void vLaunch(void) {
// Start the main task
TaskHandle_t task;
xTaskCreate(mainTask, "MainThread", 500, NULL, TASK_PRIORITY, &task);
/* Start the tasks and timer running. */
vTaskStartScheduler();
}
/***
* Main
* @return
*/
int main(void) {
// Setup serial over USB and give a few seconds to settle before we start
stdio_init_all();
sleep_ms(2000);
printf("GO\n");
// Start tasks and scheduler
const char *rtos_name = "FreeRTOS";
printf("Starting %s on core 0:\n", rtos_name);
vLaunch();
return 0;
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
MEMORY
XRTOS
SERIAL MONITOR
+ Open an additional monitor
Monitor Mode
View Mode
Text
Port
COM3 - USB 序列设置 (COM3)
Baud rate
115200
Line ending
None
Stop Monitoring
Task: 3 stk:32 hwm:32 hwm:968 Tmr Svc
Number of tasks 8
Task: 1 stk:1 hwm:1 hwm:204 MainThread
Task: 2 stk:30 hwm:30 hwm:233 IDLE
Task: 5 stk:1 hwm:1 hwm:83 Worker 1
Task: 6 stk:1 hwm:1 hwm:83 Worker 4
Task: 8 stk:1 hwm:1 hwm:83 Worker 2
Task: 7 stk:1 hwm:1 hwm:83 Worker 3
Task: 4 stk:2 hwm:2 hwm:87 Blink
Send as Text
Send Ctrl + E


MAKER PI PICO
micro SD
GP10 - GP15
AUDIO
GP18 - GP19
3V3 GND RUN
SWCLK
GND
SWDIO
BUZZER
GP18
RUN
GP20
GP21
GP22
ESP-01
RGB
USB
VREF
16 17 GND 18 19 20 21 GND 22 RUN 26 27 GND 28 29 3V3 23 GND Vin Vout


/**
* Main task to blink 20 external LEDs
*/
void mainTask(void *params){
// 1. 在 Stack 宣告 20 個 BlinkAgent 實例
BlinkAgent b0(LED0_PAD); BlinkAgent b1(LED1_PAD);
BlinkAgent b2(LED2_PAD); BlinkAgent b3(LED3_PAD);
BlinkAgent b4(LED4_PAD); BlinkAgent b5(LED5_PAD);
BlinkAgent b6(LED6_PAD); BlinkAgent b7(LED7_PAD);
BlinkAgent b8(LED8_PAD); BlinkAgent b9(LED9_PAD);
BlinkAgent b10(LED10_PAD); BlinkAgent b11(LED11_PAD);
BlinkAgent b12(LED12_PAD); BlinkAgent b13(LED13_PAD);
BlinkAgent b14(LED14_PAD); BlinkAgent b15(LED15_PAD);
BlinkAgent b16(LED16_PAD); BlinkAgent b17(LED17_PAD);
BlinkAgent b18(LED18_PAD); BlinkAgent b19(LED19_PAD);
printf("Starting 20 Blink Tasks...\n");
// 2. 逐一啟動任務
b0.start("B0", TASK_PRIORITY); b1.start("B1", TASK_PRIORITY);
b2.start("B2", TASK_PRIORITY); b3.start("B3", TASK_PRIORITY);
b4.start("B4", TASK_PRIORITY); b5.start("B5", TASK_PRIORITY);
b6.start("B6", TASK_PRIORITY); b7.start("B7", TASK_PRIORITY);
b8.start("B8", TASK_PRIORITY); b9.start("B9", TASK_PRIORITY);
b10.start("B10", TASK_PRIORITY); b11.start("B11", TASK_PRIORITY);
b12.start("B12", TASK_PRIORITY); b13.start("B13", TASK_PRIORITY);
b14.start("B14", TASK_PRIORITY); b15.start("B15", TASK_PRIORITY);
b16.start("B16", TASK_PRIORITY); b17.start("B17", TASK_PRIORITY);
b18.start("B18", TASK_PRIORITY); b19.start("B19", TASK_PRIORITY);
while (true) {
New MicroPython Project
New Rust Project
New Zephyr Project
Import Project
New Project From Example
Manage Components
Project
Debug Project
Compile Project
Run Project (USB)
Flash Project (SWD)
Configure CMake
Clean CMake
Switch Build Type Release
Switch SDK 2.2.0
Switch Board N/A
Debug Layout
Documentation
Hardware APIs
High Level APIs
Networking Libraries
Runtime Infrastructure
Serial Monitor
PROBLEMS 19 OUTPUT DEBUG CONSOLE TERMINAL PORTS MEMORY XRTOS SERIAL MONITOR
Open an additional monitor
Monitor Mode Serial View Mode Text Port COM3 - USB 序列裝置 (COM3) Baud rate 115200 Line ending None
Stop Monitoring
Task: B18 | Prio: 1 | HighWater: 87
Task: B19 | Prio: 1 | HighWater: 87
Task: B0 | Prio: 1 | HighWater: 87
Task: B1 | Prio: 1 | HighWater: 87
Task: Tmr Svc | Prio: 31 | HighWater: 968
Free Heap: 106232 bytes
Type in a message to send to the serial port.
Send as Text
Ln 117, Col 2 Tab Size: 4 UTF-8 LF

README

Chap 4 使得系統可以處理更多的任務，並且在 RP2040 上測試 FreeRTOS 的多任務處理能力。增加更多任務並觀察系統的邊界，有助於了解 RP2040 在處理大量並行任務時的性能表現

Chap 5 的重點是 **計數型信號量** 的應用，它有效地限制了同時運行的任務數量，確保了系統資源的合理分配。通過使用信號量，我們成功地同步了多個 worker 任務，並且在任務運行的過程中能夠實時檢查系統資源的使用情況。